FUSED INPUT, PROTECTION AND 5 V SUPPLY

C1
VIN_PROT 1 2 GND
10u 63V EEEFK1J100P

C2
VIN_PROT 1 2 GND
2.2u 100V X7R

C3
VIN_PROT 1 2 GND
2.2u 100V X7R

U1
GND 1 — GND
VIN_PROT 2 — VIN
UVLO 3 — EN_UVLO
RON 4 — RON
FB 5 — FB
PGOOD 6 — PGOOD
BST 7 — BST
SW 8 — SW
EP_GND 9 — GND
LM5164QDDARQ1

L1
SW 1 2 VS_LOGIC
68u MSS1246T-683MLC

R1
RON 1 2 GND
41.2k

R2
VIN_PROT 1 2 UVLO
1M

R3
UVLO 1 2 GND
221k

R4
VS_LOGIC 1 2 FB
158k

R5
FB 1 2 GND
49.9k

R6
SW 1 2 RIPPLE
150k

C4
BST 1 2 SW
2.2n 50V X7R

C5
RIPPLE 1 2 VS_LOGIC
3.3n 50V X7R

C6
RIPPLE 1 2 FB
180p C0G

C7
VS_LOGIC 1 2 GND
22u 25V X7R

C8
VS_LOGIC 1 2 GND
22u 25V X7R

R7
VS_LOGIC 1 2 PGOOD
100k

F1
VEH_ACC 1 2 ACC_FUSED
3A MINI MAIN / LED

F2
LED_PDS 1 2 VIN_PROT
1A MINI LOGIC

U4
DGATE — DGATE
ACC_FUSED — VHS
OV_FEED — SW
OV_SENSE — CV
POWER_MID — CP/UVLO
GND 7 — GND
HGATE 8 — HGATE
OUT 9 — LED_POS
VS 10 — POWER_MID
CAP 11 — CAP_PUMP
C 12 — POWER_MID
BTN 13
LM74800-Q1

D1
ACC_FUSED 1 2 GND
TPSMB33CA

Q5
ACC_FUSED — 1
ACC_FUSED — 2
ACC_FUSED — 3
4 — DGATE
5 — POWER_MID
BUK9Y15-100E

Q6
LED_PDS — 1
LED_PDS — 2
LED_PDS — 3
4 — HGATE
5 — POWER_MID
BUK9Y15-100E

R21
OV_FEED 1 2 OV_SENSE
46.4k 0.1%

R22
OV_SENSE 1 2 GND
4.02k 0.1%

R23
HGATE 1 2 INRUSH
100R

C13
ACC_FUSED 1 2 GND
100n 100V X7R

C14
POWER_MID 1 2 GND
100n 100V X7R

C15
CAP_PUMP 1 2 POWER_MID
100n 50V X7R

C16
INRUSH 1 2 GND
10n 100V X7R

C17
LED_PDS 1 2 GND
47u 50V

PWR_FLAG
GND

PWR_FLAG
VEH_ACC

PWR_FLAG
ACC_FUSED

PWR_FLAG
LED_POS

PWR_FLAG
VIN_PROT

PWR_FLAG
VS_LOGIC

VEHICLE INPUT AND CONTROLLER STACK

H1
PWM_B — 1 — 5
2 — 6
PWM_G — 3 — 7
PWM_R — 4 — 8 — PWM_B
IOL_NOMINAL

H2
ENABLE_REQ — 1 — 6
2 — 7 — GND
3 — 8
4 — 9
5 — 10
IOH_NOMINAL

H3
1 — 5 — ESP_5V
2 — 6 — GND
3 — 7 — GND
ESP_3V3 — 4 — 8
POWER_NOMINAL

H4
PWM_B — 1 — 4
2 — GPIO5 — 5 — TAIL_ADC
3 — GPIO4 — 6 — DIM_ADC
AD_NOMINAL

TP5
VS_LOGIC 1
VS_LOGIC

TP6
GND 1
GND

TP7
PGOOD 1
PGOOD

TP8
ENABLE_REQ 1
ENABLE_REQ

J1
GND 1 — IR1-5 — IR1-1 — 3 — TAIL_12V
VEH_ACC 2 — IR1-6 — IR1-2 — 4 — DIM_RETURN
TOYOTA IR1 IN

J2
LED_RGBW — 1
LED_RGBW — 2
LED_RGBW — 3
4 — LED_POS
5 — LED_W_NEG
RGBW OUT

JP1
VS_LOGIC 1 2 ESP_5V
5V RUN SHUNT

MH1
M3 CASE SUPPORT

MH2
M3 CASE SUPPORT

MH3
M3 CASE SUPPORT

MH4
M3 CASE SUPPORT

ONE COMMON-POSITIVE RGBW ZONE

U2
ENABLE_REQ — ENA — OUTB — 5 — DRV_G
PWM_G — INA — VDD — 6 — VS_LOGIC
GND 3 — GND — OUTA — 7 — DRV_R
PWM_G — INB — ENB — 8 — ENABLE_REQ
UCC27524AQDRQ1

U3
ENABLE_REQ — ENA — OUTB — 5 — DRV_W
PWM_B — INA — VDD — 6 — VS_LOGIC
GND 3 — GND — OUTA — 7 — DRV_B
PWM_W — INB — ENB — 8 — ENABLE_REQ
UCC27524AQDRQ1

C9
VS_LOGIC 1 2 GND
100n 25V X7R

C11
VS_LOGIC 1 2 GND
100n 25V X7R

C10
VS_LOGIC 1 2 GND
1u 25V X7R

C12
VS_LOGIC 1 2 GND
1u 25V X7R

R8
ENABLE_REQ 1 2 GND
2.2k

Q1
GND 1 — S1
GND 2 — S2
GND 3 — S3
G — 4 — GATE_R
D_MB — 5 — LED_R_NEG
BUK9Y15-60E

R9
DRV_R 1 2 GATE_R
100

R13
GATE_R 1 2 GND
10k

R17
PWM_R 1 2 GND
100k

D3
LED_RGBW 1 2 GND
SMBJ24CA

Q2
GND 1 — S1
GND 2 — S2
GND 3 — S3
G — 4 — GATE_G
D_MB — 5 — LED_G_NEG
BUK9Y15-60E

R10
DRV_G 1 2 GATE_G
100

R14
GATE_G 1 2 GND
10k

R18
PWM_G 1 2 GND
100k

D4
LED_RGBW 1 2 GND
SMBJ24CA

Q3
GND 1 — S1
GND 2 — S2
GND 3 — S3
G — 4 — GATE_B
D_MB — 5 — LED_B_NEG
BUK9Y15-60E

R11
DRV_B 1 2 GATE_B
100

R15
GATE_B 1 2 GND
10k

R19
PWM_B 1 2 GND
100k

D5
LED_RGBW 1 2 GND
SMBJ24CA

Q4
GND 1 — S1
GND 2 — S2
GND 3 — S3
G — 4 — GATE_W
D_MB — 5 — LED_W_NEG
BUK9Y15-60E

R12
DRV_W 1 2 GATE_W
100

EVALUATION HARDWARE / VERIFY VEHICLE FEED, LED PINOUT AND USB ISOLATION

R16
GATE_W 1 2 GND
10k

R20
PWM_W 1 2 GND
100k

D6
LED_RGBW 1 2 GND
SMBJ24CA

TOYOTA IR1 ILLUMINATION SENSE

U5
SENSE_START — SRC — D3 — 9 — GND
GND 2 — S1A — S3A — 10 — GND
TAIL_CLAMP — S1B — S3B — 11 — GND
TAIL_IN — D1 — D4 — 12 — GND
GND 3 — S2A — S4A — 13 — GND
DIM_CLAMP — S2B — S4B — 14 — GND
DIM_IN — D2 — EN_N — 15 — GND
GND 8 — GND — VDD — 16 — ESP_3V3
SN3257-Q1

U8
SENSE_START — TRGT_N — VDD — 3 — ESP_3V3
GND 2 — GND
TLV809E 2.93V

R24
TAIL_IN 1 2 TAIL_SERIES
49.9k 0.1%

R25
TAIL_SERIES 1 2 TAIL_CLAMP
49.9k 0.1%

R28
TAIL_CLAMP 1 2 GND
22k 0.1%

R30
TAIL_IN 1 2 TAIL_ADC
1k 0.1%

R32
TAIL_ADC 1 2 GND
22k 0.1%

U6
TAIL_CLAMP — A — A/NC — 3 — GND
GND 2 — K
LM4040 2.5V

D7
TAIL_CLAMP 1 A 2 GND
BAT54H-Q

C19
TAIL_ADC 1 2 GND
100n 50V

R26
DIM_RETURN 1 2 DIM_SERIES
49.9k 0.1%

R27
DIM_SERIES 1 2 DIM_CLAMP
49.9k 0.1%

R29
DIM_CLAMP 1 2 GND
22k 0.1%

R31
DIM_IN 1 2 DIM_ADC
1k 0.1%

R33
DIM_ADC 1 2 GND
22k 0.1%

U7
DIM_CLAMP — A — A/NC — 3 — GND
GND 2 — K
LM4040 2.5V

D8
DIM_CLAMP 1 A 2 GND
BAT54H-Q

C20
DIM_ADC 1 2 GND
100n 50V

R16
GATE_W 1 2 GND
10k

C18
ESP_3V3 1 2 GND
100n 50V

C21
ESP_3V3 1 2 GND
100n 50V

Sheet: /
File: vhos_rgbw_revd.kicad_sch
Title: 4Runner RGBW shield with Toyota illumination inputs
Size: A1 | Date: | **Rev: D**
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